Payout Replay

Know who actually got paid.

Non-custodial payout execution and reconciliation for DAOs and Web3 teams. Every line of the approved sheet is proven against finalized Portaldot V3 state, and only what the chain proves unpaid is ever paid again.

Portaldot Hacker House 2026 · Ecosystem DApps · payout-replay.vercel.app

THE PROBLEM

A submitted payment is not a paid one.

01

Timeouts leave rows unknown

A wallet prompt or RPC call times out halfway through a run. The sheet can no longer say whether that line went out.

02

Guessing pays twice or never

Send the line again and a contributor may be paid twice. Skip it and they may never be paid. Both happen.

03

Reconciliation is manual

Someone matches explorer pages to spreadsheet rows by hand after every run, across every chain and token.

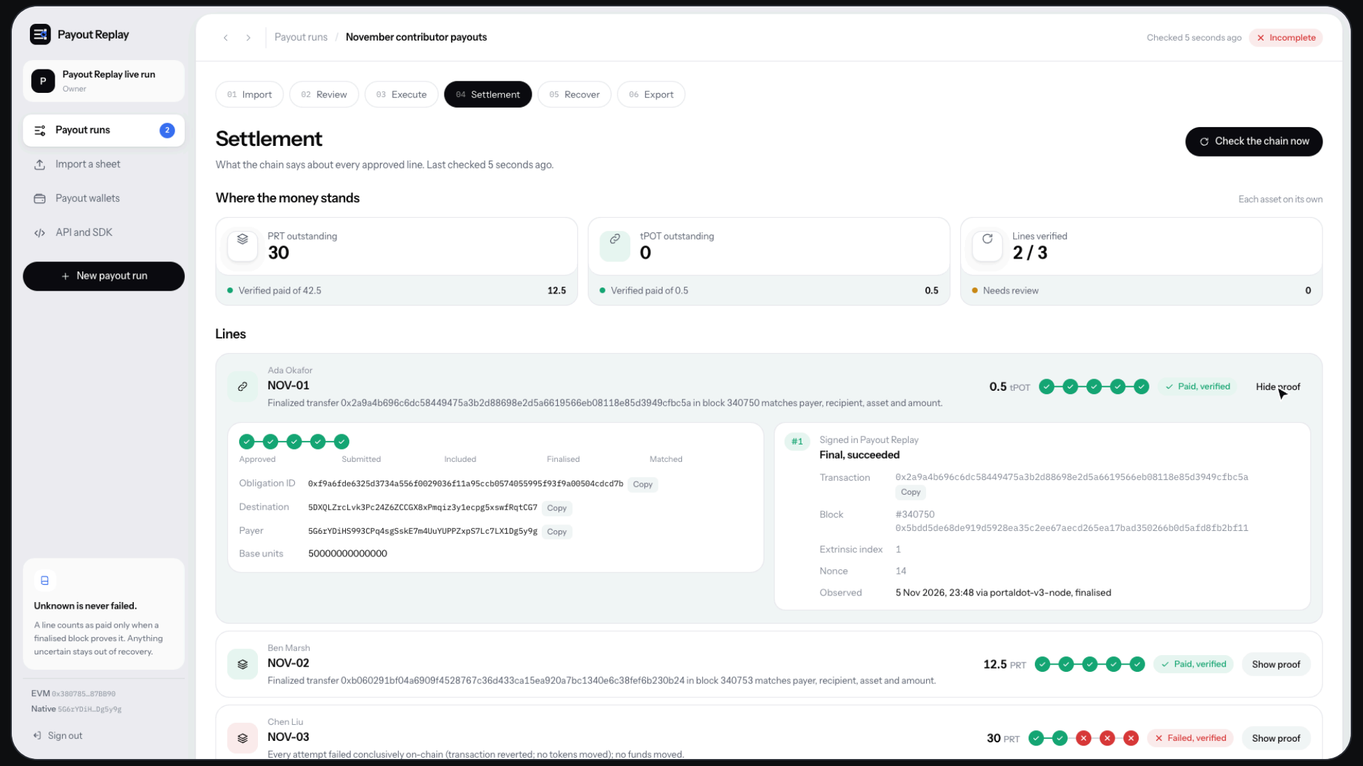
Approved, submitted and paid are three different states. Most tools treat them as one.

THE SOLUTION

Every line, proven against the chain.

Payout Replay takes the sheet a team already approved, has the treasury wallet sign each payment, and marks a line paid only when a finalized block shows the exact payer, recipient, asset and amount.

Uncertain lines stay in review. Recovery only offers lines the chain proves unpaid.



Live app on Portaldot V3 testnet, run pr_ash15eez9sabt55gex8f · 3

HOW IT WORKS

From approved sheet to proven settlement

01

Import

Every address, asset and decimal checked line by line.

02

Freeze

A reviewer approves one SHA-256 fingerprint of the sheet.

03

Sign

The treasury wallet signs. The app never holds a key.

04

Verify

Paid only when a finalized block proves the exact transfer.

05

Recover

Re-pay only lines the chain proves unpaid. Export the proof.

Unknown never becomes failed. A line with a lost wallet response stays in review until its 64-block mortal era closes or its nonce is used, so the chain itself decides when it is safe to pay again.

WHY PORTALDOT

One treasury, two runtimes, one sheet

Native and EVM in one run

V3 runs a Substrate runtime and an EVM side by side. A team pays tPOT natively and ERC-20s through pallet-revive, and reconciles both in one sheet.

GRANDPA finality

Deterministic finality means “paid” is a fact, not a probability. A line is marked paid at or below the finalized head.

Mortal eras

Native transfers are signed with a 64-block era. Once it closes unfinalized, the payment can never land, so “not paid” becomes provable.

Account mapping

One operator, one identity. H160 to native mapping is checked against ReviveApi.account_id, and the Portaldot Extension signs the native side.

PRODUCT AND TECHNOLOGY

A fail-closed reconciliation engine

Core engine

Strict CSV import, canonical manifest and SHA-256 digest, keccak obligation IDs, reconciliation and recovery planning.

Chain adapters

Native: Balances events attributed by extrinsic index. EVM: executor and Transfer events paired, nonce-based failure proof.

PayoutExecutor

Solidity on pallet-revive. Refuses a second payment for the same obligation ID and checks the recipient's balance delta.

API and worker

Hono and Postgres with immutable records, idempotent writes, wallet-signature auth and roles, plus a background reconciler.

Operator app

Next.js. Import, review, signing queue, live settlement with proof, recovery and export.

Open SDK and API

TypeScript SDK, REST API, signed webhooks and an accounting sync example. MIT licensed.

PROOF IT WORKS

Deployed and tested on V3 testnet

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live scenario checks passed

Evidence	Value
PayoutExecutor	`0x09a46d341cBB557b84311aAA8A95cB40992464e1`
Native tPOT payout	`0x97169b56…7b5a65eeb2 · block 340484`
EVM token payout	`0x26a9f1e8…614492863b · block 340491`
Duplicate replay	`0x89ea48a6…4baefb001f · reverted`
Browser run, prod app	Native and EVM signing, revert, recovery, export all passed
Automated tests	90 core · 20 Foundry incl. fuzz · 13 API end-to-end · 7 EVM · 6 SDK

EVM chain 420420777 · full hashes in docs/real-run-results.md · 7

MARKET

Treasuries pay people on-chain, at scale

$25.1B

held in DAO treasuries, $22B of it liquid. DeepDAO, March 2023.

3% → 9.6%

of blockchain workers paid in crypto in one year, 63% of them in USDC. Pantera Blockchain Compensation Survey 2024, 1,600+ responses.

$1.3B

processed by one crypto invoicing and payroll tool by January 2026, 88% in stablecoins. Request Finance, self-reported.

The errors are public too: Arbitrum DAO paid three delegates twice in 2024, and each had to send the ARB back.

Sources: Cointelegraph/DeepDAO 2023-03-30 · panteracapital.com · request.finance · Arbitrum DAO forum 2025-02-12 · 8

COMPETITION

Others send the batch. We prove it settled.

Capability	Safe CSV Airdrop	Disperse	Request Finance	Coinshift	Payout Replay
Batch payouts, non-custodial	Yes	Yes	Yes	Yes	**Yes**
Each line proven at finality	Not advertised	Not advertised	Not advertised	Not advertised	**Yes**
Unknown held, never re-paid	Not advertised	Not advertised	Not advertised	Not advertised	**Yes**
On-chain duplicate guard	Not advertised	Not advertised	Not advertised	Not advertised	**Yes**
Substrate native and EVM	EVM only	EVM only	None found	None found	**Yes**
Runs on Portaldot V3	None found	None found	None found	None found	**Yes**

From public product pages and docs, checked November 2026. “Not advertised” means no such claim was found. · 9

BUSINESS MODEL

Open core, paid hosting

Open source

Free

Engine, adapters, contract and SDK under MIT. Self-host the whole stack.

Team

$99 / month

Hosted app for a treasury paying contributors every month. Up to 500 lines a month.

Treasury

$399 / month

Unlimited lines, webhooks, accounting exports, multi-approver review and priority support.

Foundations and grant programs that pay many recipients get an annual licence. Stage: pre-revenue, pricing is proposed and will be set with the first pilots.

TRACTION AND ROADMAP

Built in the Hacker House. Next, mainnet.

Done

Product on V3 testnet

Engine, contracts, API, app and SDK. Deployed, 8 of 8 live checks, production browser run.

Next

Mainnet and audit

Move to Portaldot mainnet, external audit of PayoutExecutor, TLS edge for the API.

Then

Pilot treasuries

Three Portaldot ecosystem teams running monthly payouts. Multi-approver review, Assets pallet tokens.

Later

Beyond one chain

Accounting integrations and cross-chain payouts using Portaldot’s multi-chain capabilities.

TEAM

Built and maintained by one engineer



Tim · winsznx

Founder · full-stack and blockchain engineer · Nigeria

Ships production Web3 products end to end: Solidity and Foundry, Substrate and polkadot.js, TypeScript backends, Next.js frontends. Builds payment and treasury tooling.

`github.com/winsznx · winszn.xyz · x.com/winsznx`

THE ASK

Back the move to mainnet

Mainnet launch support

Architecture review of PayoutExecutor and revive gas, then mainnet deployment.

Building subsidy

Hosting for the API and database, and an external contract audit.

Pilot introductions

Ecosystem teams and grant programs that pay contributors every month.

Ecosystem listing

Listed as the payout tool for Portaldot treasuries, with joint onboarding.